P.8 Transformations of Functions

Name _____ Period ____ Date _____

Pre-Calculus P.8 · Student sheet

I can…

- describe vertical and horizontal shifts.
- describe reflections.
- describe stretches and shrinks and write a transformed function.

Vocabulary · fill in the missing word

Rigid transformation — a shift or _____: the graph changes position but not shape

Nonrigid transformation — a _____ or shrink: the graph changes shape

Part 1 · Slider lab

Graph $y = a|x - h| + k$ with sliders. Move one at a time.

Slider	What it does
h (positive, then negative)	
k	
$a > 1$	
$0 < a < 1$	
$a < 0$	

Part 2 · Notes and examples (fill in each note, then work the example)

Shifts: $y = f(x) + c$ moves up c; $y = f(x - c)$ moves _____ c.

Example 1

Describe $g(x) = (x + 2)^2 - 3$ compared with $f(x) = x^2$.

Reflections: $y = -f(x)$ reflects in the x-axis; $y = f(-x)$ reflects in the _____.

Example 2

Describe each compared with $f(x) = \sqrt{x}$.

a. $g(x) = -\sqrt{x}$

b. $h(x) = \sqrt{-x}$

c. $k(x) = -\sqrt{x + 2}$

Stretches and shrinks: $y = cf(x)$ is a vertical stretch ($c > 1$) or shrink ($0 < c < 1$); $y = f(cx)$ is a _____ shrink when $c > 1$.

Example 3

Describe each compared with the parent.

a. $g(x) = 3|x|$

b. $h(x) = (\frac{1}{3})|x|$

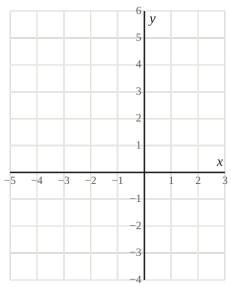
c. $k(x) = (2x)^2$

Example 4

Write the function: $y = x^3$ shifted right 4, reflected in the x-axis, and shifted up 2.

Example 5

Describe $g(x) = -2(x + 1)^2 + 5$ compared with $f(x) = x^2$, and give its vertex.



Part 3 · Partner practice

Partner 1

Describe $g(x) = \sqrt{x - 3} + 1$ compared with $f(x) = \sqrt{x}$.

Partner 2

Write the function: $y = |x|$ reflected in the x-axis, shifted left 2 and down 4.